Massive Production of Cancer Synthetic RNA-Seq Gene Expression Samples - Supplementary Materials

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1 Expression Levels Analysis

The idea of this file is to provide the remaining analysis about the performance of each generative model to learn the specific gene's behaviour.

Fig. 1 compares the real tumour samples' box plots against the box plot of the samples generated by each generative model, with the gene *BAK1* from the BRCA cohort. The GAN+MSE is located in a specific part of the real tumour box plot, which indicates overfitting. GAN+L1 contemplates a minimal part of the interquartile range, which also could suggest overfitting to a reduced range of possible values. Both VAEs did better, displaying a shape similar to the real box plot. The WGAN-GP can learn the median value, and demonstrates to learn the distribution of the genes.

Regarding *WNT2*, which is presented in Fig. 2, all the models tend to adapt to the interquartile range, except for GAN+MSE. The WGAN-GP extends the range but maintains the median value.

For *EGC* from the BRCA cohort, whose box plots are shown in Fig 3, the behaviour is similar to the previous cases, with the WGAN and VAE models achieving an acceptable representation of the interquartile range.

With *TCF7L1* from the THCA cohort, the box plots in Fig. 4 show that the VAEs tend to get the median of the distribution closer to normal samples, as they centred their generation on the third quartile. The GAN+MSE focus on the second quartile, so it prioritises the generation of downregulated samples of the gene. The GAN+L1

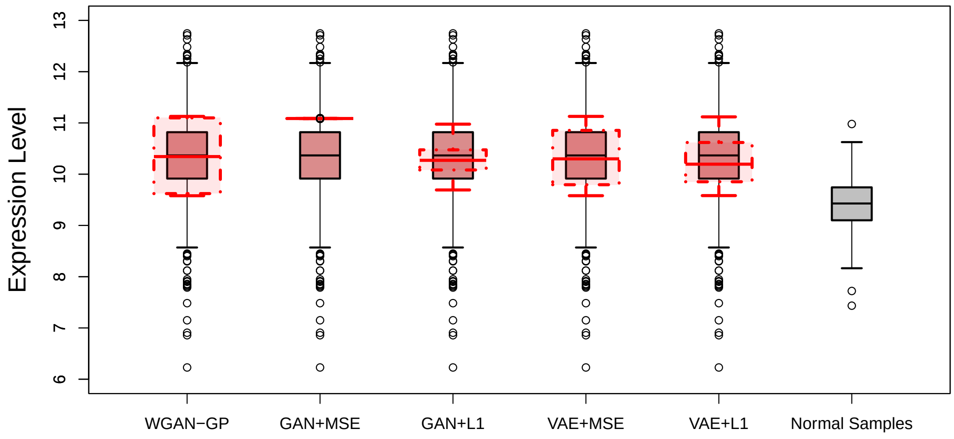

Fig. 1 Comparing the different generative models against the WGAN-GP at generating samples for *BAK1* gene from the BRCA cohort.

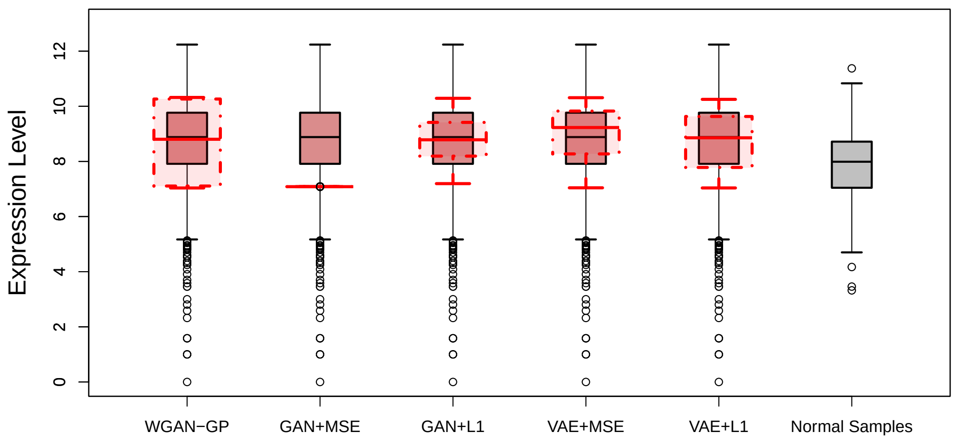

Fig. 2 Comparing the different generative models against the WGAN-GP at generating samples for *WNT2* gene from the BRCA cohort.

and the WGAN-GP are the models that better adjust to the distribution exhibited in the interquartile range of real tumour samples.

Finally, it is evident that the WGAN-GP can provide a stable performance across the genes and cohorts evaluated, resulting in an effective acquisition of the behavioural patterns of each dataset. The VAE models proved to be able to partially learn these

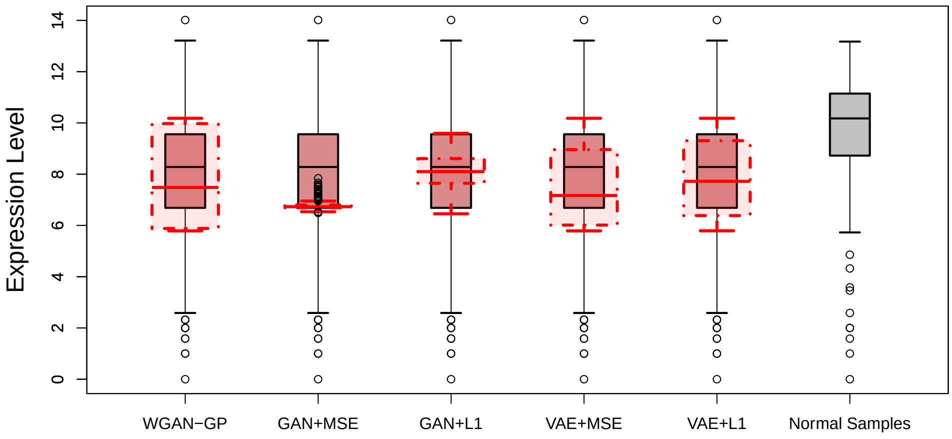

Fig. 3 Comparing the different generative models against the WGAN-GP at generating samples for *EGF* gene from the BRCA cohort.

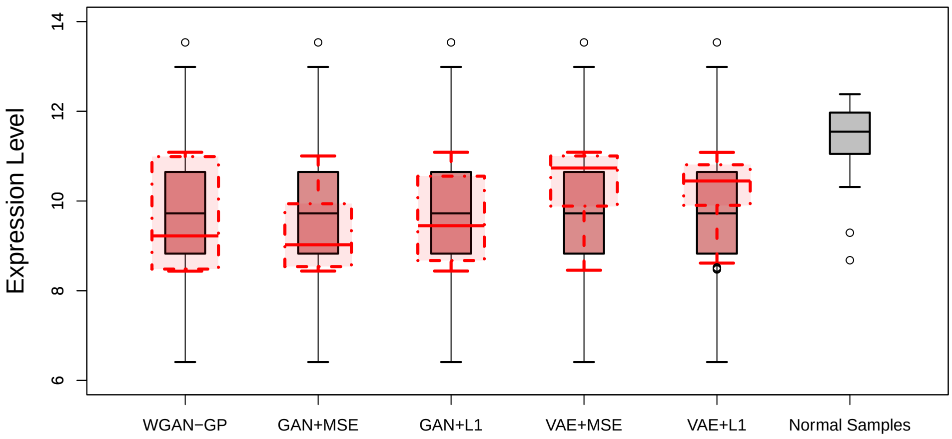

Fig. 4 Comparing the different generative models against the WGAN-GP at generating samples for *TCF7L1* gene from the THCA cohort.

patterns, achieving competitive results. The vanilla GANs were the models that struggled the most with the particularities of the datasets, showing a tendency to training overfitting.